

A Life-cycle Energy Analysis of Harnessing Solar and Wind Energy in Tall Buildings

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Abstract: This paper evaluates the life-cycle energy performance of building-integrated photovoltaics (BIPV) and small urban wind turbines (SUWTs) on tall buildings in Melbourne, Australia. Employing a concise life-cycle assessment (LCA), 6,480 scenarios are modelled to quantify average annual on-site energy generation via replacing electricity delivered by grid. Under current and future electricity mix conditions in Victoria, results show that building-integrated photovoltaics and wind turbine (BIPV-WT) systems can achieve net-positive performance (energy return-on-investment rate > 1, energy payback time < 25 years) in most cases, with a mean energy return-on-investment rate (EROI) of 2.31 and mean energy payback time (EPBT) of 14.46 years. While BIPV dominates total energy output, the shaded length of tall building emerges as the dominant factor: increases in shading linearly reduce energy return and nonlinearly extend payback times. Findings of the paper offer practical design parameters, such as limiting shaded façade to ≤ 40 % to maximise life-cycle energy yield and inform architects, planners, and policymakers on integrating hybrid renewable technologies into future high-rise developments.

Keywords: Life cycle assessment, photovoltaic panels, urban wind turbines, high-rise buildings.

1. Introduction

Despite rapid growth in renewable energy deployment, around 80 % of the primary energy continues to be derived from fossil fuels, globally. The combustion of fossil fuels drives greenhouse-gas emissions and accelerates climate change (Ritchie et al., 2024; IPCC, 2021). Buildings are responsible for over 30% of global final energy use and about 26 % of related carbon dioxide (CO₂) emissions, placing a significant share of mitigation responsibility on the built environment (IEA, 2023). In Melbourne, Australia, solar irradiance and elevated rooftop wind speeds present potentials for on-site generation of renewable energy. Tall buildings in particular offer extensive vertical surfaces for building-integrated photovoltaics (BIPV) and rooftop conditions ideal for small urban wind turbines (SUWTs). Moreover, solar output peaks during summer while urban wind speeds often rise in winter and at night (Tripanagnostopoulos et al., 2010), suggesting that a hybrid building-integrated photovoltaics and wind turbine (BIPV-WT) system could smooth the variability of energy production in a dense city context.

Recent advances have demonstrated the feasibility of façade-integrated photovoltaics in tall buildings and the potential of SUWTs in harnessing rooftop winds. Nevertheless, research on BIPV and SUWT has proceeded in parallel, with minimal cross-integration. BIPV research has focused on standalone energy yields (Zhang et al., 2018; Shirinbakhsh & Harvey, 2023) and material impacts (Wikoff et al., 2022), while assessment on SUWT's performance in urban context has been an issue, often without life-cycle assessment (LCA) of energy (Tsionas et al., 2025). Hybrid solar-wind integration research largely focuses on large, remote grid-connected farms (Lamnatou et al., 2024), leaving a notable gap in our understanding of on-site energy production within dense cityscapes. In particular, there is no systematic LCA of BIPV-WT systems in tall buildings, nor comprehensive

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assessments of their combined life-cycle energy return and payback under urban shading and local electricity-mix condition.

This paper aims at evaluating the life-cycle energy performance of hybrid BIPV-WT systems on tall buildings in Melbourne, Australia. It focuses on the urban environment, applying a LCA to quantify on-site generation versus grid electricity displacement while considering embodied energy — the sum of all energy inputs required to produce materials and services, across their supply chain. Key variables include building geometry, climate data, and the local electricity mix. By isolating Melbourne as a case study, the paper (1) quantifies annual energy production by building-integrated photovoltaics and small urban wind turbines, (2) assesses life-cycle energy return and payback time of building-integrated photovoltaics and small urban wind turbines based on local electricity mix of Melbourne, and (3) identifies critical design parameters for maximizing life cycle energy gains. The findings aim to guide architects, engineers, and planners in integrating hybrid renewables into future high-rise developments for more sustainable and resilient urban energy infrastructure, in Melbourne and beyond.

2. Methods

2.1. Overall research design

The study adopts a computational-analytical framework to evaluate the life-cycle energy performance of BIPV-WT systems in tall buildings within the urban environment of Melbourne. A parametric approach is employed to generate a suite of tall buildings with varying geometric parameters, reflecting different urban morphologies, based on which python-driven simulations integrate building geometries, local climate data and technical specifications to model BIPV-WT energy generation.

In this study, the BIPV-WT systems studied in tall building is a mini-grid system (Ortega-Arriaga et al., 2021) as shown in Figure 1, ensuring continuous energy supply to power building operations. If the capability of BIPV-WT system exceeds the demand of powering building operations, the surplus electricity could be fed into the urban grid, offsetting grid reliance for other buildings as well.

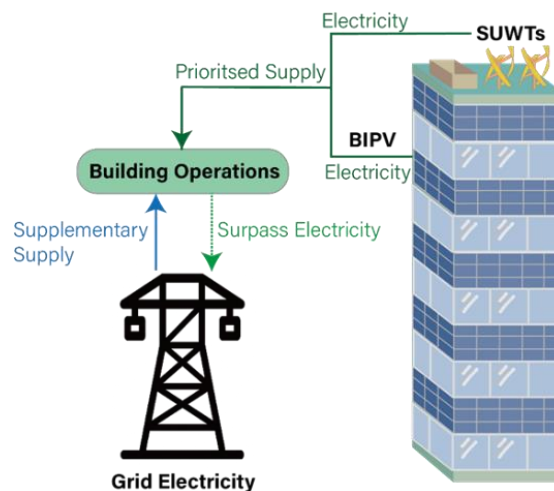


Figure 1: Hybrid building-integrated photovoltaic (BIPV) and small urban wind turbine (SUWT) systems

Table 1 summarises the primary data which results in 6,480 scenarios that are assessed in the study. According to CTBUH (2023), tall buildings in Melbourne are concentrated in urban centres. The building geometry is parameterised by height, width, and shading conditions, with 120 combinations representing a realistic spectrum of tall buildings in Melbourne. To compare the BIPV-WT energy generation against the ongoing grid decarbonisation progress, current and future predicted state-specific primary energy conversion factors (PEF_{el}) that account for electricity mixes are derived from government agency data (Department of Climate Change, Energy, the Environment and Water, 2024; Clean Energy Regulator, 2023). The technical specifications and material composition of BIPV and SUWTs are derived from manufacturer data and literature. Because of limited data availability, we include only a single BIPV type and a single SUWT model in Table 1, while the generalised Eq. (1) and Eq. (2) in Section 2.2 are intended to accommodate alternative types and their

combinations in future studies. Embodied energy and greenhouse gas emissions values are sourced from the EPiC database (Crawford et al., 2022; Crawford et al., 2024).

Table 1: Value and justification of key parameters

Data Category	Key Parameters	Value Range	Justification
Building Geometry	Height	[50, 60, 70, 80, 90, 100, 110, 120, 130, 140, 150, 160, 170, 180, 190, 200] m	According to CTBUH Data Handbook 2024, height of registered tall buildings of Melbourne ranges from 32m to 317m.
	Width	[20, 30, 40] m	For tall buildings, 25-50m square edge is the most common floor plan (Taranath, 2017). Refer to CTBUH, the model should also be slender enough to give the appearance of a tall building.
	Shaded length (0-80%)	[0, 20, 40, 60, 80] %	Intervals represent common urban obstruction scenarios: 0% (unobstructed), 20-40% (low-rise surroundings), 60-80% (dense high-rise areas).
Climate Conditions	Average annual rooftop wind speed (m/s)	Based on Melbourne city and building heights	Global Wind Atlas (Davis et al., 2023) provides validated wind speed data that varies with building height.
	Average annual solar irradiance (kWh/m ²)	Based on Melbourne city	Solar irradiance of all orientations (North, East, West and South) is simulated using Ladybug Tools (2025) and Grasshopper.
Technology Performance	BIPV: Rated power (W), Efficiency (%)	1 BIPV type	A monocrystalline silicon BIPV panel is selected.
	SUWT: Power curves	1 turbine model	A small urban wind turbine included in the previous study is chosen.
	Life span of BIPV-WT system	25 years	The life span of BIPV panel is 30 years while that of SUWT is around 25 years. A conservative 25-year life span is studied for the whole system.
Material Composition	Embodied energy (MJ/FU), Embodied GHG emissions (kgCO ₂ e/FU)	Based on BIPV and WT types	Material composition and weight of BIPV panel is estimated based on the specifications. The detailed breakdowns of SUWT are gained from the previous study. Most embodied coefficients are from the EPiC database (Crawford et al., 2022; Crawford et al., 2024). For materials that are not included in the database: factors for monocrystalline Si are gained from Wikoff et al. (2022); factors for Ethylene-vinyl acetate are assessed based on its manufacturing process.
Energy Factors	Primary energy conversion factor for electricity (GJ ^{PRIMARY} /GJ ^{DELIVERED})	(1) Current PEF _{el} ; (2) PEF _{el} declines linearly till 2050; (3) PEF _{el} declines logarithmically till 2050 (Table 4)	The current and declined values of PEF _{el} reflect state-specific electricity mix and potential grid decarbonisation progress in the future. Factor value is assessed by energy statistics published by Department of Climate Change, Energy, the Environment and Water (2024) and Clean Energy Regulator (2023).
<u>Total Number of Studied Scenarios</u>		16 heights × 3 widths × 5 shaded lengths × 3 PEF _{el} levels × 3 uncertainties on solar energy output × 3 uncertainties on wind energy output = <u>6,480 scenarios</u> .	

2.2. Quantifying the generation of renewable energy

The annual output of the BIPV-WT systems is modelled through two parallel streams. A decay factor is applied in both streams to examine the performance degradation over the life span of each system.

The average annual BIPVs energy output is calculated using Eq. (1), where the building model is represented by a combination of height (H), width (W) and shaded length (SL). To fully exploit the solar potential of tall buildings, the BIPV panels are uniformly installed facing all four orientations at a certain BIPV-to-wall ratio (e.g., 80%). The shaded length (SL) is applied (0–80%) to the number of panels, indicating the amount of unshaded BIPVs. The total number of BIPVs in the building will be used in Eq. (5). Additionally, there are 2 scalars applied to indicate the loss of panel's output overtime. Although Sepúlveda-Oviedo (2025) reports 60–70% yield losses under combined environmental factors, we adopt a conservative environmental loss scalar Φ^E ($= 0.8$) to cover limitations not captured for BIPV on tall buildings. The degradation of panel's performance is represented by a scalar Φ_{ar} ($= 0.9$) as output for the ELEGANTE 210 Wp BIPV panel decreases approximately linearly from 98% in year 0 to 86.8% at its end-of-life (by 88%).

$$BIPVE_{p,b} = \sum_{or=1}^{OR} PVE_{or} \times RP_{or,p} \times \underbrace{\left[\frac{H_b \times W_b \times BR}{RP_p / E_p} \times (1 - SL_b) \right]}_{\text{Unshaded number of panels per orientation}} \times \Phi^E \times \Phi_{ar} \quad (1)$$

Where:

$BIPVE_{p,b}$ is average annual energy generation of BIPV type p in building b , in kWh; or is the certain orientation (i.e., North, South, East or West) of building b ; H_b is building height of b ; W_b is building width of b ; PVE_{or} is average annual solar irradiance at or , in kWh/m²; $RP_{or,p}$ is total rated power of BIPV array at or of BIPV p , in kWp; BR is BIPV-to-wall ratio; E_p is efficiency of p ; SL_b is shaded length of b ; Φ^E ($= 0.8$) is a scalar representing environmental losses due to temperature decrease, dust, etc.; Φ_{ar} ($= 0.9$) is a scalar representing the decrease in panel output over time.

Meanwhile, the SUWTs of the system are positioned along the rooftop perimeter to maximise wind capture from multidirectional urban airflow. A spacing of three times turbine diameter is maintained to minimise wake effects. Thereby, as shown by Eq. (2), while height (H) of the combined building model determines the wind speed, the total wind energy generation is restricted by width (W) as well.

$$WTE_{t,b} = WTP_t(s_b) \times \underbrace{\left[\frac{W_b}{3 \times D_t} \right] \times 4}_{\text{Total number of wind turbines}} \times Hrs \times \Phi_{ar} \quad (2)$$

Where:

$WTE_{t,b}$ is average annual energy generation of SUWT type t in building b , in kWh; $WTP(s)$ is power curve of SUWT type t in kW; s_b is average annual rooftop wind speed of b , in m/s; W_b is building width of b ; D_t is the turbine diameter of t in m; Hrs ($= 8,760$) is the number of turbine working hours per year; Φ_{ar} ($= 0.9$) is a scalar representing the decrease in turbine output over time.

Once the annual output of BIPVs and SUWTs are calculated respectively, the total annual output of the BIPV-WT systems in the building model is quantified as per Eq. (3):

$$ATE_{p,t,b} = BIPVE_{p,b} + WTE_{t,b} \quad (3)$$

Where:

$ATE_{p,t,b}$ is average annual total energy generation of BIPV type p and SUWT type t in building b , in kWh; $BIPVE_{p,b}$ is average annual energy generation of BIPV type p in building b , in kWh; $WTE_{t,b}$ is average annual energy generation of SUWT type t in building b , in kWh.

2.3. Quantifying energy savings through grid displacement

The on-site renewable energy is prioritised in the building model to support operations, which displaces grid electricity. Eq. (4) quantifies the amount of energy saved on-site by primary energy conversion factor for electricity (PEF_{el}) based on the electricity mix in Victoria state:

$$ATESO = ATE \times 0.0036 \times PEF_{el} \quad (4)$$

Where:

$ATESO$ is the average annual energy saved on-site by BIPV-WT system in GJ; ATE is average annual total energy generation of BIPV-WT system in kWh; PEF_{el} is the average primary energy conversion factor for electricity of Victoria state, in $\text{GJ}^{\text{PRIMARY}}/\text{GJ}^{\text{DELIVERED}}$.

2.4. Hybrid life-cycle energy assessment

For the purpose of implementing the system in tall buildings, the scope of LCA focuses on the embodied environmental flows across the supply chain of BIPVs and SUWTs materials. Process analysis is the most widely used LCA approach for materials, involving inputs and outputs data of environmental flows in each process (Finnveden et al., 2009). However, according to the EPiC database published by Crawford et al. (2024), more than 60% of the environmental flow effects are not accounted for in the process data.

To fill this gap in the supply chain, Eq. (5) below is a process-based hybrid LCA model applied to quantify the embodied energy of the BIPV-WT system. The model uses hybrid LCA data from the Australian EPiC database (Crawford et al., 2022; Crawford et al., 2024), which provides hybrid embodied energy flow coefficients for a wide range of construction materials. For materials not covered in the EPiC database (Crawford et al., 2022; Crawford et al., 2024), the production processes are individually investigated based on relevant literature. To minimise truncation errors, hybrid coefficients (Φ_{hy}) are applied to the process-based values extracted from literature, aiming to better capture the environmental flows across the full supply chain. In the sensitivity analysis, a further safety approach is conducted to ensure the robustness of the results and to avoid underestimating the embodied energy of the overall BIPV-WT systems.

$$E_E = \sum_{s=1}^S \sum_{te=1}^{TE} \sum_{m=1}^M Q_{s,te,m} \times EEC_{s,te,m} \times \Phi_m \quad (5)$$

Where:

E_E is the total embodied energy of BIPV-WT system in GJ; S is the total number of installed systems of technology type te (e.g., building-integrated photovoltaics or small urban wind turbines); TE is the total number of technologies; M is the total number of materials in each system s ; $Q_{s,te,m}$ is the quantity of material m in system s in technology type te of system s , in functional units (e.g., m^2 of Monocrystalline Si); $EEC_{s,te,m}$ is hybrid embodied energy coefficient, for material m in system s in technology type te in unit GJ per functional unit of m (e.g., GJ/m^2); Φ_m is a hybrid factor for m that covers remaining gaps of process-based data in the supply chain.

2.5. Performance metrics and analysis

The assessment of the performance of the BIPV-WT systems focuses on energy savings within its life span.

As per Eq. (6), the return-on-investment rate (ROI) represents the ratio of the total energy generated by the system over its life span to the embodied energy. A higher ROI indicates a more efficient and sustainable energy system, capable of generating significantly more energy than it consumes over its life cycle.

$$EROI = \frac{ATESO \times LS}{E_E} \quad (6)$$

Where:

$EROI$ is energy return-on-investment rate; $ATESO$ is the average annual energy saved on-site by BIPV-WT system in GJ; LS (= 25 yrs) is the life span of BIPV-WT system; E_E is the total embodied energy of BIPV-WT system in GJ.

As per Eq. (7), the payback time (PBT) measures the time required for the system to compensate for its own embodied energy flows.

$$EPBT = \frac{E_E}{ATESO} \quad (7)$$

Where:

$EPBT$ is energy payback time in year; E_E is the total embodied energy of BIPV-WT system in GJ; $ATESO$ is the average annual energy saved on-site by BIPV-WT system in GJ.

In the context of this study, high EROI values would support the capability of using tall buildings equipped with the BIPV-WT systems as “urban power banks”, while payback times shorter than the life span suggest a transition to net positive performance before the system’s end-of-life. These indicators illustrate whether the BIPV-WT systems could contribute to a broader decarbonisation and urban resilience objectives.

2.6. Sensitivity analysis

Given the complexity and interdependence of the datasets involved in this study, a sensitivity analysis is conducted to assess the robustness of the BIPV-WT system performance under varying assumptions. The implementation of BIPV and SUWT technologies in tall buildings relies heavily on site-specific factors, including urban morphology, microclimatic conditions, and material sourcing, all of which can significantly influence the accuracy of life-cycle environmental performance estimations.

In urban environments, the available solar resource for façades may be substantially reduced due to shading effects from neighbouring buildings (Şirin et al., 2023). Similarly, the wind energy potential can be constrained by the turbulent and obstructed nature of urban wind flows, which directly affects the performance of SUWTs (Tsionas et al., 2025). Additionally, the current application of SUWTs in dense urban areas remains under-researched, and their aerodynamic interactions with high-rise structures are still not fully understood (Tsionas et al., 2025). These uncertainties can lead to either overestimation or underestimation of the annual energy output.

The sensitivity analysis ensures that the core results of BIPV-WT system in urban tall buildings are not overly reliant on a narrow set of assumptions and can better reflect the diversity of real-world applications. Key variables, tested scenarios and justifications are summarised in Table 2.

Table 2: Parameters of sensitivity analysis

Parameter	Scenarios	Justification
Average annual solar irradiation (kWh/m ²)	±20%	To simulate unforeseen impact of urban environment on solar resource
Average annual wind energy generation (kWh)	±30%	To reflect variability in urban wind conditions (e.g., turbulence, wind directions) and limited research on SUWTs

3. Results and findings

As summarised in Table 3, the maximum annual energy outputs per total façade area of BIPV-WT systems is 68.47 kWh/m² in Melbourne, with the peak value occurring at a building height of 190m, a width of 30m and 0% shaded length. Under the net-positive criteria (EROI > 1 and EPBT < 25 years) for energy generation, it shows favourable performance: the mean EROI is 2.31 (median = 2.32), and the mean EPBT of 14.46 years (median = 10.78 years) lies well below the 25-year threshold. Although a minority of outliers approach the limits (minimum EROI = 0.77; maximum EPBT = 32.4 years), the majority of scenarios meet the net-positive condition.

Under the same building width and shaded length conditions, the energy output per total façade area of BIPV-WT systems increases slightly with building height. However, the difference is minimal, for example, increasing the height from 50m to 200m results in only a 1 kWh/(m²·a) rise in annual energy output. It reflects that parameters such as shaded length and regional electricity mix (characterised by PEF_{el}) govern BIPV-WT's life-cycle performance.

Table 3: Maximum annual energy output per facade area (each orientation), building conditions for maximum energy output, life-cycle energy and greenhouse gas emissions performance

City	Max. Annual Energy Output per façade area [kWh/(m ² ·a)]		Max. Energy Conditions	Performance Metrics	EROI (1)	EPBT (years)
Melbourne	68.47		Height: 190m Width: 30m Shaded: 0%	Range	0.77-3.86	6.5-32.4
	N: 91.73	E: 74.96		Average	2.31	14.46
				Median	2.32	10.78
	S: 34.35	W: 72.82		Standard Deviation	1.06	8.76

3.1. Share of total energy output

The result represents that the performance of hybrid system is fundamentally driven by the BIPV system. For 30m-wide building models in Melbourne, BIPV share is shown to account for at least 87% of the total energy output, while at 60m height without shading, it peaks at 98.9%. Wind turbine output remains marginal around 13% even at 180m-200m height when the building is shaded by 80% and with high wind speeds at the top. This finding aligns with the maximum annual energy output per façade area at each orientation (North, East, South and West) presented in Table 3, as values of energy output follow the same directional pattern as the simulated average annual solar irradiance in Melbourne (1131 kWh/m² at Northern façade, 920 kWh/m² at Eastern façade, 409 kWh/m² at Southern façade and 893 kWh/m² at Western façade).

The shaded length of the building could dramatically diminish BIPV performance. When the shaded length remains below 20%, increasing building height from 50m to 200m reduces BIPV contribution by merely 1-2%. Under more severe conditions, for instance, at 80% shaded length, BIPV supplies 94% of total energy generation at 50m height, but its share declines to 88% at 200m. Only at this scale could wind turbine output become a necessary complement to offset the energy deficit.

Since BIPV could not generate electricity at night and its output might fluctuate under cloudy conditions, further research should investigate wind turbine performance at night and during low-irradiance periods. Additionally, beyond the BIPV–WT system, incorporating gravity batteries into tall buildings might help ensure consistent electricity supply (Emrani et al., 2024).

3.2. Significance of shaded length

Figures 2 present correlations between shaded length and metrics (EROI and EPBT), showing that shaded length emerges as the dominant parameter governing life-cycle energy performance and the influence of building height on the performance is negligible. While metrics differ a little when building height varies, EROI exhibits a linear decline by 0.75 as shaded length increases by 20% (Figure 2 | Left). In contrast, EPBT rises in a nonlinear manner (Figure 2 | Right). When the shaded length rises from 60% to 80%, the PBT rises drastically from 16 years to 32 years, approximately.

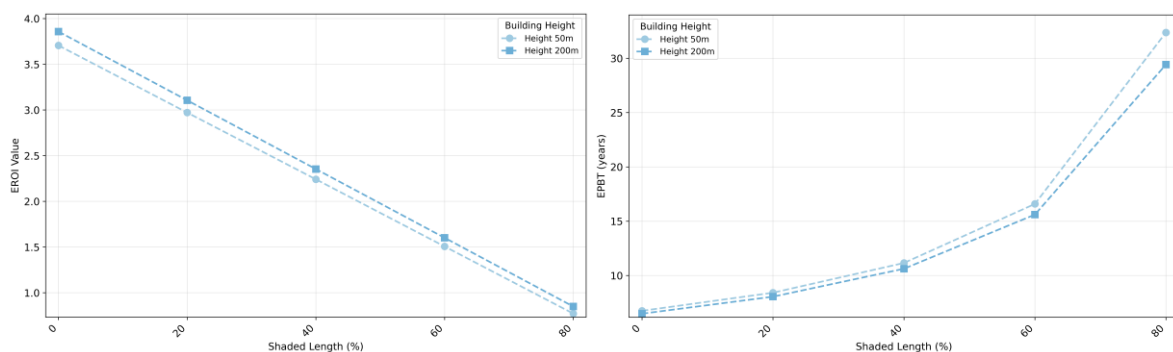


Figure 2: Left: Correlation between EROI and shaded length of BIPV-WT systems in 200m-high and 50m-high building models (width = 30m) in Melbourne under current electricity mix conditions, Right: Correlation between EPBT and shaded length of BIPV-WT systems in 200m-high and 50m-high building models (width = 30m) in Melbourne under current electricity mix conditions

3.3. Significance of the electricity mix

The electricity mix in this paper refers to the electricity supplied by grid that the BIPV–WT system could displace on site. A cross-validation process is applied to primary energy and energy generation figures provided by Department of Climate Change, Energy, the Environment and Water (2024) as there are discrepancies of fuel type across different energy datasets. Wherever a fuel type listed in dataset was missing, its value is estimated by applying known generation efficiencies (Feng, 2023) to the closest available data from either database. Current PEF_{el} value is calculated once these gaps are filled, assuming that the transmission and distribution loss of energy is 10% in Victoria.

It is assumed that the state-specific PEF_{el} value can be used for Melbourne. Table 4 presents the current PEF_{el} value and future average PEF_{el} scenarios for the state of Victoria, assuming that 90 % primary energy would be renewable by 2050. These values enable assessment on BIPV-WT system's performance as Australia's grids become progressively cleaner.

Table 4: Scenarios and values of primary energy conversion factor for electricity based on the electricity mix in Victoria state

Scenarios	Average PEF_{el} until 2050 ($GJ^{PRIMARY}/GJ^{DELIVERED}$)
Current	2.9
Linear value decline	2.1
Logarithmic value decline	1.7

Figure 4 plots EROI (vertical axis) against EPBT (horizontal axis) for 405 simulated building models (height 160–200m, 40 % shaded) in Melbourne, by declining PEF_{el} values. The figure incorporates uncertainties in both solar energy ($\pm 20\%$) and wind energy ($\pm 30\%$) generation, which are the main sources of the error bars. It is clear that at a restricted shaded length, BIPV-WT systems in tall buildings could achieve net-positive energy generation ($EROI > 1$ and $EPBT < 25$ years) in Melbourne during Australia's decarbonisation progress.

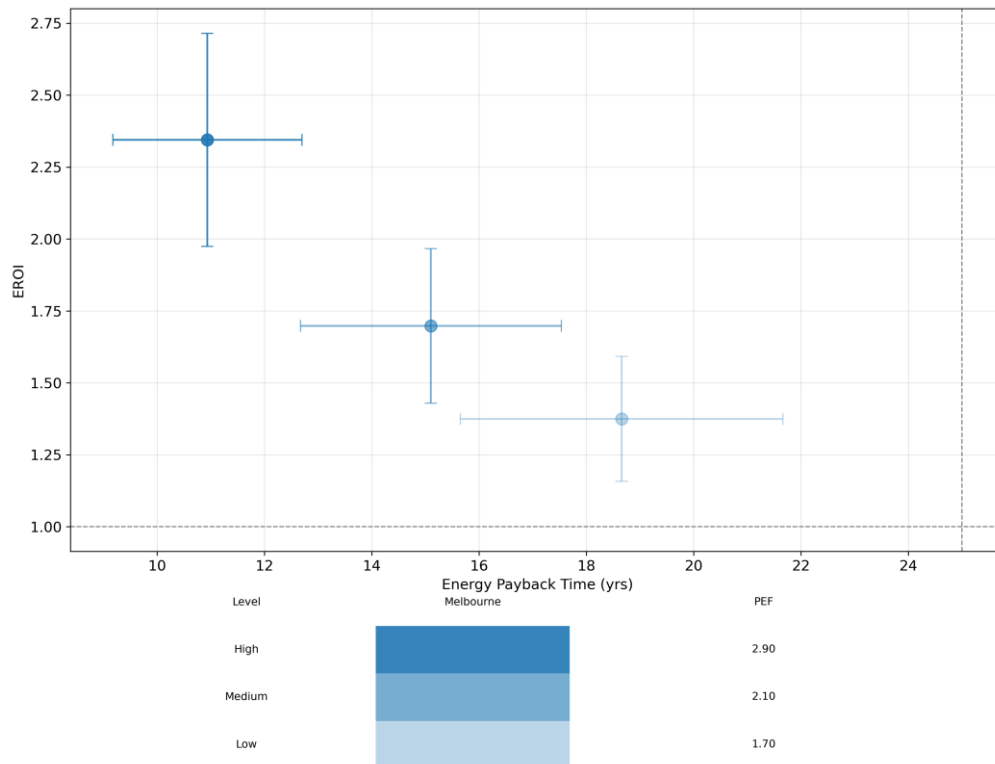


Figure 3: Energy return-on-investment and energy payback time of BIPV-WT systems under current and future PEF_{el} in 160-200m high and 40% shaded building models with uncertainties on solar or wind energy outputs

The scatter plot highlights the impact of the electricity mix on system performance. During the decarbonisation of the regional electricity mix, EROI of BIPV-WT systems drops largely, from 2.3 to 1.7 and 1.4, while EPBT extends by almost 68%, from 11 years to 18.5 years. It also shows that EROI tends to be more sensitive to energy output fluctuations when the PEF_{el} value is high, whereas EPBT is more significantly affected by such uncertainties when PEF_{el} value is low. At the lowest PEF_{el} value ($1.70 GJ^{PRIMARY}/GJ^{DELIVERED}$), the standard deviation of EPBT is around 6 years, and the longest payback time reaches 21.5 years.

4. Discussion and conclusion

4.1. Contribution

This study has quantified the life-cycle energy performance of hybrid BIPV–WT systems installed on tall buildings in Melbourne. It also evaluates the regional electricity mix, obtaining both current and projected PEF_{el} values. With these values, the research advances prior research on hybrid solar–wind systems by modelling on-site electricity generation in urban environments, integrating LCA to assess energy ROI and payback time (PBT), and performing a sensitivity analysis to evaluate future feasibility.

By measuring maximum energy output per square metre of façade, this study shows that BIPV–WT systems in Melbourne can generate 68.47 kWh/(m²·a) under current electricity-mix conditions. Notably, the total energy output is contributed by BIPV by around 90%. While Shirinbakhsh and Harvey (2023) found that covering all available façade area with BIPV alone could not achieve net-zero energy supply in tall buildings, our results reveal that the marginal contributions from wind turbines require further investigation on alternative integration of on-site renewables and real-time performance assessment.

The LCA results for BIPV–WT systems on tall buildings indicate that net-positive performance ($ROI > 1$, $PBT < 25$ years) for energy generation is achievable in Melbourne, under current and future decarbonised electricity mixes. It suggests that BIPV–WT systems could make a meaningful contribution towards national Australian targets of 100 % renewable-energy target by 2050. However, the life cycle performance of the system is highly sensitive to the shaded-length of the building and the primary energy conversion factor of the electricity mix. During the transition to cleaner grids, the less shaded a building is, the better its BIPV–WT performance is over its lifespan. While developing tall buildings with BIPV–WT systems, it is recommended to control the building shaded length no larger than 40% to reach EROIs larger than 3 and EPBT shorter than 10 years. This will also help with solar access to the inhabitants of these buildings.

4.2. Limitations and future research

While this study provides a LCA on on-site hybrid sustainable technologies under urban conditions, several limitations suggest directions for future research.

Firstly, the scope is restricted to energy metrics in a single city. As the paper has quantified only life-cycle energy return (EROI) and payback time (EPBT), future studies should extend the LCA to include greenhouse gas emissions or other environmental impacts to provide a more holistic sustainability appraisal. Also, comparative analyses across multiple climatic zones and urban contexts would help test the transferability of design guidelines.

Moreover, the sensitivity analysis in this study is limited in its consideration of technology types. As noted by Wikoff et al. (2022), the embodied environmental flows of BIPV modules can vary significantly depending on the photovoltaic cell material used. To better understand how technology choices influence life-cycle performance, future research should incorporate a broader range of BIPV and SUWT technologies and conduct sensitivity analyses on their embodied environmental flows.

4.3. Conclusion

This paper demonstrates that integrating BIPV and SUWTs on Melbourne's tall buildings can achieve net-positive life-cycle energy performance (mean EROI = 2.31; mean EPBT = 14.46 years) in most modelled scenarios. Shaded length emerges as the primary determinant of energy yield, while the local electricity mix modulates overall benefits. These findings underscore the value of hybrid renewable systems in dense cities. By establishing practical design limits, such as maintaining façade shading below 40 %, the paper provides clear guidance for architects and policymakers to enhance urban high-rise sustainability.

5. Acknowledgement

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